

TermPlanMT: Graph-Grounded Terminology Planning for French–English Regulatory Medical Document Translation

When Ontology Grounding Helps and When It Does Not

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Abstract

Terminology consistency is a critical quality dimension in the machine translation of regulatory medical documents, where mistranslated adverse event terms carry direct patient-safety consequences. We present **TermPlanMT**, a graph-augmented terminology-planning framework evaluated on a 126-segment French–English corpus drawn from a European Medicines Agency (EMA) Summary of Product Characteristics (SmPC). Six systems form an ablation ladder isolating the contributions of NER recall, MedDRA ontology grounding, and constrained decoding. Our central finding is that Concept Coverage Rate (CCR), the fraction of source spans grounded in a MedDRA ontology graph, is the binding constraint on terminology-aware MT: at 39% CCR, graph-free systems (S1–S2) outperform all graph-augmented systems on fluency—S1 leads on BLEU, S2 on chrF and Hypothesis–Reference Agreement (HRA)—while improving NER quality yields ≈ 7 BLEU gains for graph-augmented Mistral systems (S3–S5–Mistral) but not for NLLB or S2. HTM scores the certified human reference at 0.15–0.18, unchanged by full MedDRA alias expansion, confirming a structural mismatch between MedDRA nomenclature and professional translation practice. Ambiguous grounding causes the planning stage to abstain from locking critical terms, explaining why graph-augmented systems produce lower cross-document consistency than graph-free baselines.

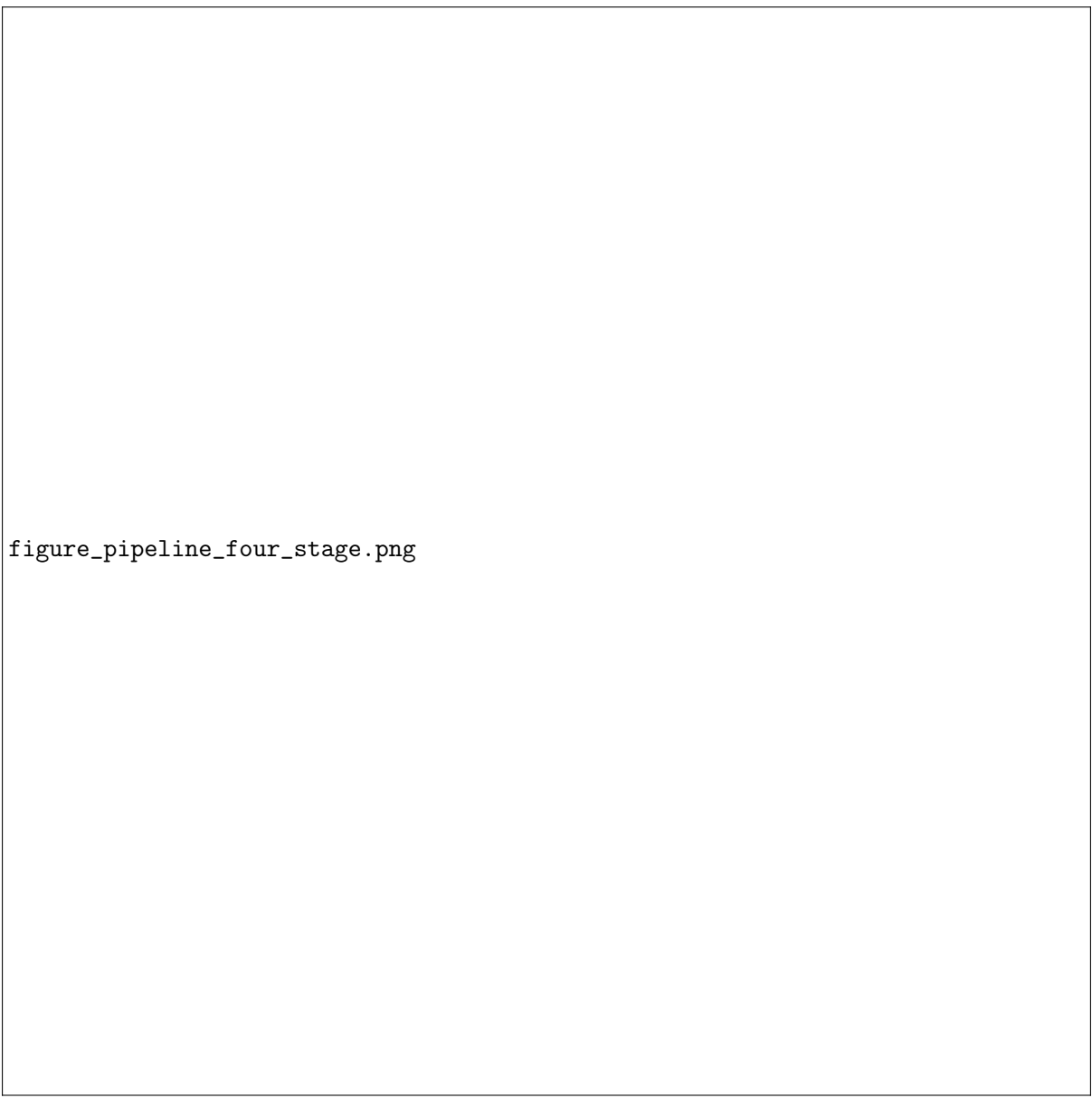
receive one identical English rendering across an entire document—a requirement that affects pharmacovigilance signal detection and patient safety (?). Neural MT handles medical prose fluently but is structurally indifferent to cross-segment consistency, producing term drift (same source term, multiple renderings), concept flattening (Lowest Level Term (LLT)-level terms collapsed to hypernyms), and coverage gaps (multi-token spans outside NER). MedDRA’s five-level hierarchy (Table ??) provides validated English labels that are an attractive grounding resource—if the grounding pipeline can cover enough of the source text.

This paper makes the following contributions:

1. **A six-system ablation ladder** that isolates the individual contributions of NER recall, MedDRA ontology grounding via Neo4j, and constrained decoding enforcement to terminology-aware MT quality.
2. **Three novel evaluation metrics** — HTM, rHTM, and HRA — that measure terminology alignment at the ontology level rather than surface string overlap, with a diagnostic showing $rHTM \approx 0.15\text{--}0.18$ for certified human references.
3. **A CCR bottleneck analysis** demonstrating that NER extraction quality, not retrieval or decoding strategy, is the primary constraint on graph-augmented MT performance.
4. **A mechanistic consistency analysis** explaining the paradox in which graph-augmented systems produce lower cross-document term consistency than graph-free baselines, traced to the planning stage’s abstain-on-ambiguity policy.
5. **Reproducible artefacts** including per-segment HTM scores, bootstrap confidence intervals,

1 Introduction

EMA Summaries of Product Characteristics (SmPCs) require a single adverse-event concept to



figure_pipeline_four_stage.png

Figure 1: Four-stage TermPlanMT pipeline (NER → grounding → planning → decoding).

and qualitative drift analysis exported as CSV and Markdown.

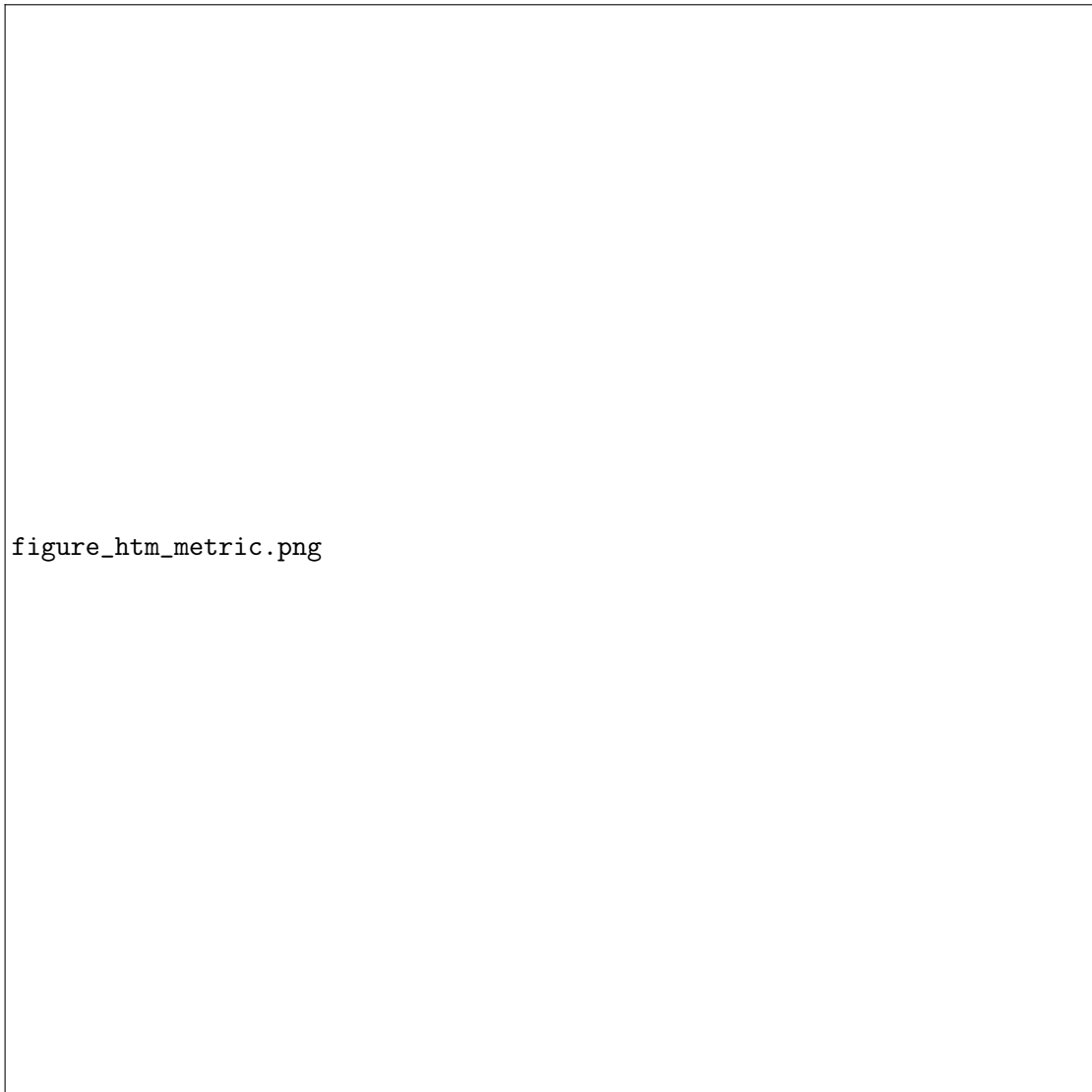
2 Related Work

Lexically constrained decoding (??) and terminology-aware training (?) enforce surface-level term appearance but ignore ontological hierarchy and do not address candidate selection when multiple levels are valid. ? show that feeding long context directly into NMT degrades output quality due to attention mechanism limitations, motivating our pre-translation planning stage: rather than exposing the full document at decoding time, we distil relevant terminology into a compact, structured prompt. ? release BioMistral, the medically adapted LLM used for NER here.

? identify biomedical named entities broadly—diseases, drugs, and procedures—as a persistent failure mode in biomedical MT. Our work narrows this to the specific subtype of adverse event (AE) terminology as defined by the MedDRA ontology, a distinction with direct regulatory consequences given that AE terminology is legally standardised across EMA submissions. GraphRAG-style retrieval (?) into LLM prompts provides the template for our S3 system, adapting knowledge-grounded MT (?) to a Neo4j MedDRA backend.

3 Dataset

Corpus.. Section 4.8 (Undesirable Effects) of the KEYTRUDA (pembrolizumab) SmPC (?): 126 aligned French–English segment pairs after ex-



figure_htm_metric.png

cluding segment 48_028 (a dense frequency table whose n-gram overlap distorts corpus metrics). Coverage: $\approx 4,400$ words; all three failure modes at high density.

Level	Abbr.	Example
L1	SOC	Respiratory disorders
L2	HLGT	Pulmonary conditions
L3	HLT	Pneumonitis
L4	PT	Pembro. pneumonitis
L5	LLT	Immune-med. pneumonitis

Table 1: MedDRA hierarchy. Adverse-event mentions in clinical text map almost exclusively to L5 ($\approx 96\%$).

NER conditions as a CCR perturbation..

Rather than comparing two NER systems, we use them as a controlled perturbation of CCR within the same pipeline. **Baseline NER** (ner_biollm):

BioMistral-7B zero-shot; CCR = 39.0%, 184 grounded spans across 472 extracted (126-segment evaluation set). This condition exhibits a systematic artefact—the model generates *hypothyroïdie* for 92/127 segments regardless of content—partially contaminating consistency measurements. **Fine-tuned (higher-quality) NER** (ner_biollm_ft): BioMistral-7B fine-tuned on a hierarchical MedDRA SFT dataset; 211 grounded spans across 580 extracted (36.4% CCR) with cleaner boundaries and no hallucinated terms. Graph-augmented systems benefit substantially under this condition, confirming that *effective* span quality (clean boundaries, usable grounding)—not raw graph retrieval—is the binding constraint once contamination is controlled.

4 System Architecture

Six systems form an ablation ladder (Table ??). S1–S5 share the same four-stage pipeline: French

NER → MedDRA grounding via Neo4j → global terminology lock → MT decoding. S6 is a glossary oracle ablation that replaces the MedDRA graph with a curated FR→EN glossary extracted from the reference translations.

ID	Model	Graph	Enforce.
S1	NLLB-200 600M	No	None
S2	Mistral-7B (doc)	No	None
S3	Mistral + GraphRAG	Yes	None
S4	S3 + reranking	Yes	Soft
S5	NLLB/Mistral + logit	Yes	Hard
S6	NLLB + glossary	Oracle	Hard

Table 2: Ablation ladder. S1 = NMT lower bound; S2 = LLM with full document context; S3 is the key novel condition; S6 isolates term-source quality from graph retrieval.

Terminology planning.. Before translation, `compute_global_locks()` scores each candidate English rendering c :

$$\text{Score}(c) = \alpha \text{Ctx}(c) + \beta \text{Nbr}(c) - \gamma \text{HP}(c) \quad (1)$$

where Ctx is cosine similarity to source segments containing the term, Nbr is similarity to adjacent MedDRA concept renderings, and $\text{HP} = |\hat{d} - d_{\text{src}}|$ penalises wrong specificity level ($\alpha=0.5$, $\beta=0.3$, $\gamma=0.2$). When a term matches *multiple* MedDRA concepts, no lock is committed (§??).

The abstain-on-ambiguity policy (line 4) is a deliberate safety choice: when a French term maps to multiple MedDRA concepts with equal confidence (e.g. *pneumopathie inflammatoire* maps to both PNEUMONITIS and INTERSTITIAL LUNG DISEASE), forcing a lock risks propagating the wrong concept across the entire document. The cost of this conservatism is reduced cross-document consistency, since unlocked terms are resolved independently at each segment by the decoder. We discuss this consistency paradox in Section ??.

S5 — Logit Boosting with Hard Fallback.. S5 applies a two-layer decoding strategy. First, a soft logit bias (PhraseLogitBoost, $\delta=1.75$ for

Algorithm 1 Global Terminology Planning (`compute_global_locks`)

Require: Segments $\mathcal{S}$, graph $\mathcal{G}$, $\alpha=0.5$, $\beta=0.3$, $\gamma=0.2$
Ensure: Lock map $\mathcal{L}$: $\text{fr_term} \rightarrow \text{en_rendering}$

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1:  $\mathcal{L} \leftarrow \{\}$ 
2: for each unique French NER term  $w$  in  $\mathcal{S}$  do
3:    $c^* \leftarrow \mathcal{G}.\text{ground}(w)$   $\triangleright$  Exact match, then RapidFuzz  $\geq 90\%$ 
4:   if  $c^*$  is None or  $\mathcal{G}.\text{is\_ambiguous}(w)$  then
5:     continue  $\triangleright$  Abstain: wrong lock propagates document-wide
6:   end if
7:    $\mathcal{C} \leftarrow \{c^*.\text{name}\} \cup \mathcal{G}.\text{neighbours}(c^*)$ 
8:   for each candidate  $r \in \mathcal{C}$  do
9:      $\text{Ctx}(r) \leftarrow \cos(\text{emb}(\text{segs}_w), \text{emb}(r))$ 
10:     $\text{Nbr}(r) \leftarrow \cos(\text{emb}(r), \text{emb}(\mathcal{G}.\text{nbrs}))$ 
11:     $\text{HP}(r) \leftarrow |\text{level}(r) - \text{level}(c^*)|$ 
12:     $\text{Score}(r) \leftarrow \alpha \text{Ctx}(r) + \beta \text{Nbr}(r) - \gamma \text{HP}(r)$ 
13:   end for
14:    $\mathcal{L}[w] \leftarrow \arg \max_{r \in \mathcal{C}} \text{Score}(r)$ 
15: end for
16: return  $\mathcal{L}$ 

```

NLLB, $\delta=1.25$ for Mistral) increases the probability mass of tokens belonging to locked MedDRA renderings without preventing the model from generating alternative phrasings if contextually motivated. Second, for NLLB only, `force_words_ids` enforces the locked phrase as a hard constraint via constrained beam search, with automatic fallback to soft-only decoding if the constraint causes beam failure. The Mistral variant uses soft boosting only, since instruction-tuned LLMs are more sensitive to hard token forcing and exhibit greater fluency degradation under strict constraints.

5 Metrics

Metric definitions.

- **HTM** (Hierarchy-Aware Terminology Metric): per grounded NER span, score = 1.0 (exact MedDRA level), 0.5 (same branch), 0.0 (miss); macro-averaged.
- **rHTM** (Reference HTM): HTM applied to the certified human reference, not a system hypothesis. A dataset-level ceiling; values of 0.15–0.18 confirm translators prefer phrasings outside the MedDRA preferred-term vocabulary.

- **HRA** (Hypothesis–Reference Agreement): $1 - |\text{HTM}_{\text{hyp}} - \text{HTM}_{\text{ref}}|$, macro-averaged. Measures agreement with professional translation practice rather than raw ontology proximity.
- **HTM-A** (Alias-Expanded HTM): accepts any ancestor up to SOC. Score invariance across tiers T1–T3 confirms failures are conceptual, not surface-level.
- **CCR** (Concept Coverage Rate): grounded spans / extracted spans. Identical across S1–S6 for a given NER condition; the primary diagnostic for NER quality.

Fluency.. Corpus BLEU (?) and chrF++ (?) via SacreBLEU against one professional English reference.

HTM and HTM-A.. For each segment containing a grounded French span, the hypothesis is scored against the MedDRA graph: $s_i = 1.0$ (exact level match), 0.5 (same branch, wrong specificity), or 0.0 (different branch / absent). $\text{HTM} = \text{mean}(s_i)$. HTM-A relaxes matching across three alias tiers: T1 exact LLT; T2 adds Preferred Term (PT) parent; T3 adds all same-branch ancestors. We compute **Reference HTM (rHTM)** by applying HTM to the professional reference as hypothesis, establishing a performance ceiling.

Hyp–Ref Agreement (HRA).. Fraction of HTM-measured segments where the system’s terminology choice matches the professional reference—a measure of human-like translation independent of MedDRA correctness.

CCR and drift.. $\text{CCR} = \text{grounded}/\text{extracted}$ spans; identical across S1–S6 for a given NER condition. Term drift = $(D - 1)/(N - 1)$ for D distinct renderings across N segments; drift = 0 is perfect consistency.

6 Results

6.1 Main Results

HTM is recall-dependent on NER quality.. BLEU and chrF scores for S1 and S2 are identical across both NER conditions (Tables ?? and ??), confirming that the underlying translations did not change. However, HTM scores shift substantially: S1 HTM rises from 0.321 to 0.550 and S2 from 0.413 to 0.735 under fine-tuned NER. This is a *metric artifact*, not a translation improvement.

System	BLEU	chrF++	COMET	HTM	HRA
S1 NLLB	22.89	37.14	0.639	0.321	0.672
S2 Mistral	22.51	39.95	0.637	0.413	0.743
S3 GraphRAG	10.39	34.36	0.586	0.397	0.601
S4 rerank	9.93	34.24	0.583	0.429	0.579
S5 NLLB+boost	21.59	35.91	0.638	0.353	0.661
S5 Mist+boost	10.32	34.32	0.585	0.380	0.607
S6 NLLB+gls*	20.00	35.11	0.634	0.321	0.694
Reference	—	—	—	0.151	1.000

Table 3: Baseline NER condition (ner_biollm, CCR = 39.0%; 126 segments, 48_028 excluded). BLEU/chrF++ via SacreBLEU; COMET via wmt22-comet-da; HTM/HRA from Neo4j. *S6 is a glossary oracle (terms from reference).

System	BLEU	chrF++	COMET	HTM	HRA
S1 NLLB	22.89	37.14	0.639	0.550	0.429
S2 Mistral	22.51	39.95	0.637	0.735	0.476
S3 GraphRAG [†]	19.58	35.33	0.626	0.664	0.440
S4 rerank [†]	19.66	35.36	0.626	0.664	0.450
S5 NLLB+boost	21.37	36.06	0.637	0.578	0.403
S5 Mist+boost [†]	20.61	35.78	0.626	0.673	0.450
S6 NLLB+gls*	20.09	34.81	0.632	0.536	0.408
Reference	—	—	—	0.178	1.000

Table 4: Fine-tuned NER condition (ner_biollm_ft; CCR = 36.4%; 126 segments, 48_028 excluded). [†]Contamination-filtered (18/127 rows scored as empty hyps). *S6 uses a glossary oracle from the reference.

HTM is computed only over spans the NER model successfully extracts; a stronger NER model exposes a larger and more challenging set of terminology spans, changing both the denominator and the matched set simultaneously. Scores across NER conditions are therefore not directly comparable and should be interpreted within each condition only.

6.2 CCR is the Binding Constraint

Table ?? isolates the effect of NER quality by holding all other pipeline components fixed and comparing the same system across NER conditions.

System	Δ BLEU	Δ chrF++	95% CI Δ BLEU
S1 NLLB	± 0.00	± 0.00	[0.00, 0.00]
S2 Mistral [‡]	± 0.00	± 0.00	[0.00, 0.00]
S3 GraphRAG	+7.07	+0.97	[+1.07, +11.10]
S4 rerank	+7.42	+1.12	[+1.18, +11.53]
S5 NLLB+boost	−0.16	+0.15	[−0.98, +0.66]
S5 Mist+boost	+7.16	+1.46	[+1.10, +11.66]
S6 NLLB+gls [*]	+0.09	−0.30	—

Table 5: Paired gain (fine-tuned minus baseline NER) on 126 segments. Δ BLEU is the paired bootstrap estimate (2,000 resamples, seed 42); 95% CIs for S3–S5-Mistral exclude 0. Δ chrF++ is the corpus-level paired difference. [‡]S2 shares the same inference JSONL across conditions. ^{*}S6 CI not computed (oracle glossary).

S1 is unchanged (no graph). S2 is exactly zero: the same inference JSONL is shared across both NER conditions, so Δ BLEU is zero by construction. S3–S5-Mistral show ≈ 7 -point Δ BLEU with intervals strictly above zero, isolating the NER bottleneck on the graph path. S5-NLLB is effectively flat. The S6 oracle glossary yields negligible gain, showing that logit boosting with reference-extracted terms does not improve over the graph path once NER quality is matched.

The 95% paired bootstrap confidence intervals for the BLEU gain of fine-tuned graph-augmented Mistral systems (S3–S5-Mistral) over their baseline-NER counterparts exclude zero (see Table ??), confirming statistical significance at $p < 0.05$. Graph-free systems (S1, S2) and S5-NLLB show confidence intervals that include zero, indicating their apparent gains are not reliably distinguished from noise.

S1 leads on corpus BLEU and S2 leads on chrF and HRA in the baseline NER condition (Table ??): graph-augmented systems fall well below both because the graph covers fewer than 40% of source spans while injected MedDRA strings hurt fluency in segments where it is active. S2’s lead on HRA (0.743) reflects the human-like output of full-document Mistral context without graph interference. The contrast between zero Δ BLEU for S1, S2, and S5-NLLB versus large Δ BLEU for S3–S5-Mistral isolates the bottleneck upstream of graph retrieval. Higher NER coverage would likely close the S2/S3 fluency gap.

6.3 Missing Terms, Flattening, and Consistency

Under baseline NER, S5-Mistral achieves the lowest missing-term rate; S1-NLLB misses the

majority of grounded terms. Graph systems benefit from cleaner NER: S3’s missing-term rate falls substantially between conditions. Zero concept flattening events are observed (all grounded L5 source terms produce L5 hypotheses), partly an evaluation artefact—HTM cannot detect flattening unless upper-level aliases are provided, a limitation HTM-A addresses.

Mean cross-document modal agreement, averaged over French terms with ≥ 2 segments in `error_analysis/`, is highest for graph-free systems under baseline NER (S1: 0.993; S2: 0.984) while S3 scores 0.946; under fine-tuned NER, S3 rises to 0.976, surpassing S1 (0.970) but remaining below S2 (0.985)—so the “graph hurts consistency” pattern is clearest under noisy baseline NER and partially attenuates when NER improves, matching the missing-term and BLEU deltas.

7 The HTM–Practice Gap

7.1 Why the Reference Scores Near Zero

On the baseline NER manifest, $r\text{HTM} = 0.151$ on the professional reference (Table ??); under fine-tuned NER the reference HTM is 0.178 (Table ??) because the grounded span set differs. In both cases the score stays below several MT systems on HTM. Professional translators do not transcribe MedDRA LLT strings verbatim: the LLT *Immune-mediated pneumonitis* appears as *pneumonitis* in tabular contexts and as *immune-related pneumonitis* in descriptive contexts.

7.2 Alias Expansion Does Not Close the Gap

Tier	Accepts	Low	Hi
T1	Exact LLT	0.151	0.178
T2	T1 + PT parent	0.151	0.178
T3	T2 + branch ancestors	0.151	0.178

Table 6: $r\text{HTM}$ -A on the professional reference under three alias tiers (Low = baseline NER manifest; Hi = fine-tuned NER manifest). All tiers yield identical scores within each manifest.

The ancestor chain for *pneumopathie inflammatoire* contains *pneumonitis*, so the null T1→T3 delta is not a graph coverage gap. Professional translators use phrasings such as *immune-related pneumonitis* and *inflammatory lung disease* that lie entirely outside the MedDRA preferred-label tree at any level. The mismatch is structural: MedDRA labels are designed for pharmacovigilance coding, not regulatory prose.

HTM should therefore be interpreted as a measure of MedDRA-proximity, not translation quality. The HRA column is the complementary measure. Notably, graph-augmented systems score consistently *lower* on HRA than S2 (Table ??: S3 = 0.601 vs. S2 = 0.743; Table ??: S3 = 0.440 vs. S2 = 0.476); the graph pushes outputs toward MedDRA labels that professional translators avoid.

8 Consistency Failure Analysis

Term drift (qualitative).. *Pneumopathie inflammatoire* occurs in four baseline segments where it is grounded for S3/S4/S5-Mistral, and in eight fine-tuned segments (more NER hits for the same French string). Those systems emit two renderings—often *pneumonitis* in incidence-style sentences and *immune-mediated pneumonitis* elsewhere—yielding drift scores of 1/3 (baseline) and 1/7 (fine-tuned). S1 collapses to one rendering for this span in both conditions (drift 0). S2 is stable in baseline (drift 0) but shows non-zero drift under fine-tuned NER once the span appears in additional contexts. S3/S4/S5-Mistral thus exhibit the clearest cross-segment inconsistency for this exemplar.

Mean modal agreement (§??) shows the same pattern at corpus level: graph-free systems rank highest under baseline NER; graph-augmented systems recover partially under fine-tuned NER but do not overtake S2.

Translator language outside MedDRA.. Professional translators systematically prefer descriptive phrasings that fall outside the MedDRA preferred-term vocabulary. We identify three categories of translator deviation from ontology labels:

1. **Descriptive expansion.** The MedDRA PT PNEUMONITIS (L4) appears in the reference as “*immune-related pneumonitis*” or “*inflammatory lung disease*”—phrasings absent from the MedDRA tree at any level, scoring 0.0 under HTM regardless of alias tier (T1–T3). This is the primary cause of the low rHTM ceiling.
2. **Hypernym substitution.** Translators deliberately ascend the hierarchy, selecting an HLT or SOC label where the broader term improves readability. HTM treats this as a specificity error even when it reflects a deliberate stylistic choice.
3. **Compound paraphrase.** Long MedDRA labels (e.g. PALMAR-PLANTAR

ERYTHRODYSAESTHESIA SYNDROME) are shortened to clinical shorthand (“*hand-foot syndrome*”), which falls outside the MedDRA branch entirely. Vector HTM at $\tau=0.75$ partially recovers these cases; surface HTM misses them entirely.

This taxonomy explains both the rHTM ceiling and the HTM–BLEU divergence: systems that score well on HTM tend to use technical ontology labels that human references avoid, producing lower BLEU against the descriptive reference while appearing more ontology-compliant. The appropriate evaluation lens is therefore **HRA**, which measures agreement with professional translation practice rather than raw ontology proximity.

Table 7: Register contrast: human translators vs. MT systems for the same adverse-event source spans.

System	Register	HTM	Consistency
Human ref	Descriptive prose	0.15–0.18	Deliberate variation
S1 NLLB	Rigid hypernym	0.32–0.55	High (vocabulary artefact)
S2 Mistral	Contextual prose	0.41–0.74	Moderate
S3/S4 Graph	Technical PT label	0.40–0.66	Lower (ambiguity)
S5 Boost	Forced PT label	0.35–0.67	Stochastic
S6 Oracle	Reference-anchored	0.32–0.54	Higher

Mechanism.. *Pneumopathie inflammatoire* maps to two distinct MedDRA concepts in the string cache (one of 7,679 ambiguous normalised keys). The planning stage sets `has_planning_lock = 0` for any ambiguous term and commits no lock. S3/S4/S5-Mistral then resolve the ambiguity context-locally per segment from RAG context: incidence-focused segments attract *pneumonitis*; mechanism-focused segments attract *immune-mediated pneumonitis*. S1-NLLB produces zero drift not because it reasons about terminology, but because its fixed vocabulary has no mechanism for context-dependent selection.

This failure is independent of CCR. Even with perfect NER recall, any term matching multiple MedDRA concepts will be left unlocked, producing potential drift in all Mistral-based graph systems. The fix is *soft ambiguity resolution*: commit the top-scoring candidate as a provisional lock flagged for human review, consistent with CAT-tool workflows.

Output contamination.. In the fine-tuned NER condition, 18/127 segments per system (14.2%) contain prompt contamination: Mistral reproduced MedDRA metadata strings and instruction markers from its own system prompt. These rows were replaced with a null hypothesis for metric computation. Contamination biases raw scores in opposite directions—deflating BLEU (precision destroyed by injected strings) while inflating chrF (character overlap increased)—confirmed by S3 raw vs. filtered: BLEU 15.36→19.58, chrF 37.20→35.33. S1 and S2 are unaffected as they receive no graph context in their prompts.

9 Discussion and Limitations

The results collectively show that the TermPlanMT pipeline is architecturally sound but CCR-limited. Graph-augmented systems improve substantially when supplied with cleaner, higher-coverage NER input. The absence of BLEU movement for S1 (and near-zero change for S5-NLLB), together with large Mistral-graph Δ BLEU in Table ??, is direct evidence that the bottleneck lies upstream of graph retrieval on the graph-augmented path.

The S6 oracle glossary (terms extracted from the reference via an LLM) yields only marginal BLEU improvement over S5-NLLB (+0.09 points), suggesting that the quality of the terminology source matters less for NLLB decoding than the NER coverage upstream.

The HRA–HTM inversion—graph systems score higher on HTM yet lower on HRA—reveals a real tension: optimising toward MedDRA formal labels diverges from how certified translators actually write. For regulatory MT in practice, HRA is the more meaningful target.

Limitations.. *Single document and reference.* All results derive from one SmPC section against one professional translation, limiting generalisability. *HTM ceiling.* rHTM is low in absolute terms (0.151–0.178 across NER manifests) and invariant to alias expansion within a manifest, so

HTM is only interpretable relatively. *NER artefact.* The baseline NER condition contains hallucinated spans (*hypothyroïdie* in 92/127 segments). *Contamination.* Fine-tuned-condition Mistral-based graph systems exhibit 14.2% prompt contamination; filtered scores are used throughout. *S6 oracle.* The glossary is extracted from reference translations via Ollama; genuine oracle quality depends on extraction recall.

Boost success rate.. S5 applies logit boosting to locked MedDRA renderings at each decoding step, but we do not directly verify whether the boosted term surfaces in the final hypothesis. A term can fail to appear in two ways: the model generates a competing token with a stronger contextual signal, or *force_words_ids* triggers a beam failure and falls back to soft-only decoding. Measuring the fraction of segments where the intended locked term appears verbatim in the hypothesis—the *boost success rate*—would directly link the planning mechanism to observed HTM scores and is an important diagnostic for future work. We expect boost success rate to correlate negatively with term specificity: broad SOC-level terms (short, common) are more likely to surface than precise LLT-level multi-token phrases.

10 Conclusion

TermPlanMT demonstrates that CCR is the binding constraint on graph-augmented regulatory MT: at low coverage, graph-free systems (S1–S2, S5-NLLB) outperform all graph-augmented systems on fluency—S1 leading on BLEU, S2 on chrF and HRA—while improving NER quality yields ≈ 7 -point corpus BLEU gains for Mistral graph systems with bootstrap intervals excluding zero. Three broader findings emerge: MedDRA nomenclature and professional translation practice are structurally misaligned in ways that alias expansion cannot fix; ambiguous grounding produces predictable term drift through a mechanism fully explained by the planning stage’s abstain-on-ambiguity policy; and consistency metrics must control for system type, since NLLB’s apparent consistency is an artefact of its fixed vocabulary. The glossary oracle (S6) shows that replacing the MedDRA graph with reference-extracted terms provides negligible benefit, pointing to NER coverage rather than term-source quality as the primary bottleneck. Substantially higher CCR with cleaner span boundaries would likely be required to re-

verse the S2/S3 fluency ordering; quantifying that threshold is left to future work with broader French biomedical NER.

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